Hints: Polynomials – 7. Interpolation and finite differences

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Lagrange: the i-th basis polynomial $\prod_{j\neq i}\frac{x-x_j}{x_i-x_j}$ has degree n, equals 1 at x_i and 0 at the other nodes. Uniqueness: the difference of two interpolants has degree $\leqslant n$ and $n+1$ roots. Equivalently the linear system for the coefficients has the Vandermonde matrix, whose determinant $\prod_{i<j}(x_j-x_i)\neq 0$.

Hint 2. $\Delta x^n=(x+1)^n-x^n=nx^{n-1}+\dots$, so Δ lowers the degree by one and multiplies the leading coefficient by the degree; iterating gives $\Delta^n f=n!\,a_n$, and $\Delta^n f=0$ if $\deg f<n$.

Hint 3. Write $\Delta=E-I$ with the shift $(Ef)(x)=f(x+1)$. These commute, so $\Delta^n=\sum_k\binom{n}{k}(-1)^{n-k}E^k$, and evaluating at 0 gives $\sum_k(-1)^k\binom{n}{k}f(k)=(-1)^n\Delta^n f(0)$.

Problem 2

Hint 1. Put $q(n)=p(2n+1)$, of degree 1008, with $q(n)=F_{2n+1}$ for $0\leqslant n\leqslant 1008$. Then $\Delta^{1009}q=0$ expresses $q(1009)=p(2019)$ through the known values.

Hint 2. $q(1009)=\sum_{k=0}^{1008}(-1)^{1008-k}\binom{1009}{k}F_{2k+1}$. Use Binet $F_m=(\varphi^m-\psi^m)/\sqrt{5}$ and $\varphi^2-1=\varphi$, $\psi^2-1=\psi$: the full sum $\sum_{k=0}^{1009}(-1)^{1009-k}\binom{1009}{k}\varphi^{2k+1}=\varphi(\varphi^2-1)^{1009}=\varphi^{1010}$.

Hint 3. Subtracting the analogous ψ-sum, the full alternating sum of F_{2k+1} over $0\leqslant k\leqslant 1009$ is F_{1010}; separating the term $k=1009$ gives $p(2019)=F_{2019}-F_{1010}$, i.e. $(j,k)=(2019,1010)$.

Problem 3

(a)

Hint 1. With $x=1/k$ the condition says $P(x)=\frac{1}{x}+2x$ for infinitely many x. Clear the denominator.

Hint 2. $xP(x)-1-2x^2$ has infinitely many roots, so it is identically zero; at $x=0$ this gives $-1=0$. Answer: no.

(b)

Hint 1. Here $\frac{1}{2k+1}=\frac{x}{x+2}$ with $x=1/k$, so $(x+2)P(x)-x$ vanishes at infinitely many points; evaluate the resulting identity at $x=-2$. Answer: no.

Problem 4

Hint 1. $g(x)=1+\sum a_i x^{b_i}$ has a zero of order n at $x=1$. Apply the operator $\theta=x\frac{d}{dx}$, which acts on monomials by $\theta x^b=bx^b$.

Hint 2. $(\theta^j g)(1)=0$ for $j<n$ gives $\sum_i a_i=-1$ and $\sum_i a_i b_i^j=0$ for $1\leqslant j\leqslant n-1$, while $(\theta^n g)(1)=(-1)^n n!\,f(1)$ because only the term where all derivatives hit $(1-x)^n$ survives.

Hint 3. Let $Q(t)=\prod(t-b_i)$. Then $0=\sum_i a_iQ(b_i)=\sum_i a_ib_i^n+Q(0)\sum_i a_i=\sum_i a_ib_i^n-Q(0)$, so $\sum_i a_ib_i^n=Q(0)=(-1)^nb_1\cdots b_n$. Answer: $f(1)=\dfrac{b_1b_2\cdots b_n}{n!}$.

Problem 5

Hint 1. Differentiate $\frac{P_n}{(x^k-1)^{n+1}}$ once to get a recursion for P_{n+1}.

Hint 2. $P_{n+1} = (x^k-1)P_n' - (n+1)kx^{k-1}P_n$, so $P_{n+1}(1) = -(n+1)k\,P_n(1)$ with $P_0 = 1$. Answer: $P_n(1) = (-k)^n\,n!$.